

Otto3 operator interface guide

Otto3 is operated through **OttoPanel** — one firmware, two equivalent interfaces: the **touchscreen** on the GIGA Display Shield (no computer needed) and a **remote serial console** (9600 baud, for shell/Claude-driven sessions). Both expose the same actions; the touchscreen adds checklists and confirmation gates, the serial console skips them (the remote operator is the confirmation).

1. Touchscreen

Home → **menu** → **checklist** → **run**

- **Home** shows three categories: **CALIBRATION** (steps 1–9), **RUN** (the full sequencing protocol), **UTILITIES** (primes, rinses, dispense/aspirate-all, park, shutdown flush, and the calibration inspection tests).
- Tapping a category opens a **2-column button grid** (cell 1 is always **< BACK**).
- Tapping an action opens its **pre-run checklist**: short physical prerequisites ("RESERVOIRS FULL", "PUMP ARMED (STOP AFTER KEYPAD USE)", "VACUUM OPEN, TRAP IN LINE, 12V ON", ...). Every row must be ticked before **GO** enables. **CANCEL** returns to the menu.

The run screen

While an action runs, the display is a live dashboard:

- **Four panels** — reagent valve (current port + reagent name), sample valve (current port), vacuum valve + solenoid state, pump state with a live countdown for the current pump segment.
- **Step timer** — elapsed vs expected duration for the whole action (expected times are computed from the live constants at confirm time).
- **Cue banner** — operator instructions timed to the fluidics: calm cues ("PRIMING - NO NEED TO WATCH YET"), armed countdowns ("EYES ON DISP VALVE - FRONT ARRIVES T-11s"), and a triple full-screen **strobe at T-0** so the watch moment is unmissable. During the purge loop each needle gets its own countdown ("RED AT NEEDLE 3").
- **The red STOP column** is always present and always hot. One tap sets the abort latch: the running routine unwinds within ~100 ms (`Wait()` polls the latch in 100 ms slices; pump/valve/solenoid primitives refuse new actions), then the instrument **parks safe** — pump stopped, vacuum solenoid closed, valves to reagent 4 / sample 1 / vacuum 1.

The result screen

When an action finishes, a **DONE** screen shows the measured duration and, for calibration tests, an inspection prompt (e.g. "INSPECT: AIR SLUG AT SAME SPOT IN ALL 6 LINES?"). Tap to return to the checklist/menu.

The Step 5 dispense wizard

CALIBRATION → step 5 runs as a guided wizard: dispense pass → operator enters the measured volume on-screen → the wizard retunes `mLPumpTime` in **RAM** and offers another pass. The converged value must still be written into `OttoFns/constants.ino` by hand (no on-device persistence) — the wizard shows the number to copy.

2. Remote serial console

Open the port at **9600 baud** (or use `tools/otto_console.py`, which does everything below for you). The firmware prints the command table at boot and on `?`. One character per command, newline-terminated.

Command table

Key	Action	Notes
1 – 9	Calibration steps 1–9	checklists skipped
R	FULL RUN (the sequencing protocol)	
w / c / i	Prime wash / cleavage / incorporation, 2 min	
f	FULL RINSE (the flush cycle, ~18 min)	
d	DISPENSE 1 mL to all wells	
a	ASPIRATE all wells	
k	PARK valves (reagent 4 / sample 1 / vacuum 1)	
r	CAL: RESET LINE (wash prime out the vent)	cleans reagent path + vent ONLY
v	CAL: BUBBLE TO VALVE (static)	superseded by <code>b</code> for tuning — static parks read ~0.1 mL low
n	CAL: BUBBLE TO NEEDLE 1 (static)	
t	CAL: SPLIT TEST — full production prefix, stops after the split	
p	CAL: PURGE TEST — the ascending purge only, run right after <code>t</code>	
b	CAL: PARK TEST — <code>t</code> truncated at the park; seals the line so the bubble front freezes for reading	
m	CAL: NEXT SPLIT SLICE — one production slice (descending from 7), reseal after	6× = the full split, stepwise
u	CAL: NUDGE TO 7 — ~13 µL titration step; prints the running total since the park	

Key	Action	Notes
M	CAL: SPLIT ALL 6 — the verbatim production split loop in one step	
x	STOP / ABORT (only accepted during a run)	same latch as the red button
s	STATUS — state, elapsed, live constants	accepted any time
?	Help	

While an action runs the console answers `BUSY (RUNNING)` to everything except `x` and `s`.

Phase markers

Every action streams progress markers, so a script (or Claude) can follow a run without watching the screen:

```

RUN: CAL: SPLIT TEST      <- action accepted and started
STEP: CAL: SPLIT TEST     <- step timer started
CUE: PRIMING — NO NEED TO WATCH YET
CUE: EYES ON DISP VALVE — FRONT ARRIVES T-11s
CUE: NOW                  <- the strobe moment
DONE 183s                 <- finished, measured duration
ABORTED                   <- (instead of DONE) the stop latch fired

```

The Step-8 stepwise toolkit (`b` → `u` → `m / M`)

Decoupled bubble-calibration protocol, every advance operator-gated:

1. `d a r` — the **clean regime**. Run it before every reading (`r` alone does not clean the six sample lines).
2. `b` — park the bubble with full production fidelity, then freeze it (line sealed; the front cannot creep). Read the front against the valve.
3. `u` (repeat) — advance ~13 μL per press until the first air tip exits into line 7. The printed total is the measured reagent-line volume.
4. `M` — the production split in one step; inspect the six air slugs. Or `m` six times — the same split one line at a time, resealing between slices so every intermediate state can be read.
5. `p` — the production purge; inspect the six wash edges at the needles.

See `KNOWN-ISSUES.md` for what these tests established on 2026-08-21 and what remains open.

3. Rules that protect the hardware

- **Never flash while a run is active** — uploading reboots the board mid-motion. Confirm `s` shows IDLE first.

- **Pump arming:** after any front-panel keypad use, press the pump's STOP key once; remote control only arms after STOP. The keypad locks out while the remote contact is closed.
- **Power sequence:** GIGA on **first**, pump power off **first**. An unpowered GIGA drives the pump's start input low — the pump runs.
- **Port 6 of the reagent valve must stay dry** (no tubing): routines draw air through it. **Port 1 of the sample and vacuum valves is never plumbed** — the software parks there and the run logic requires it open.